slam-toolbox Mapping

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1. Course Content

Basic: Run the example program, build and save a grid map with the slam-toolbox mapping algorithm.

Advanced: Understand slam-toolbox's configuration parameters

2. slam-toolbox Introduction

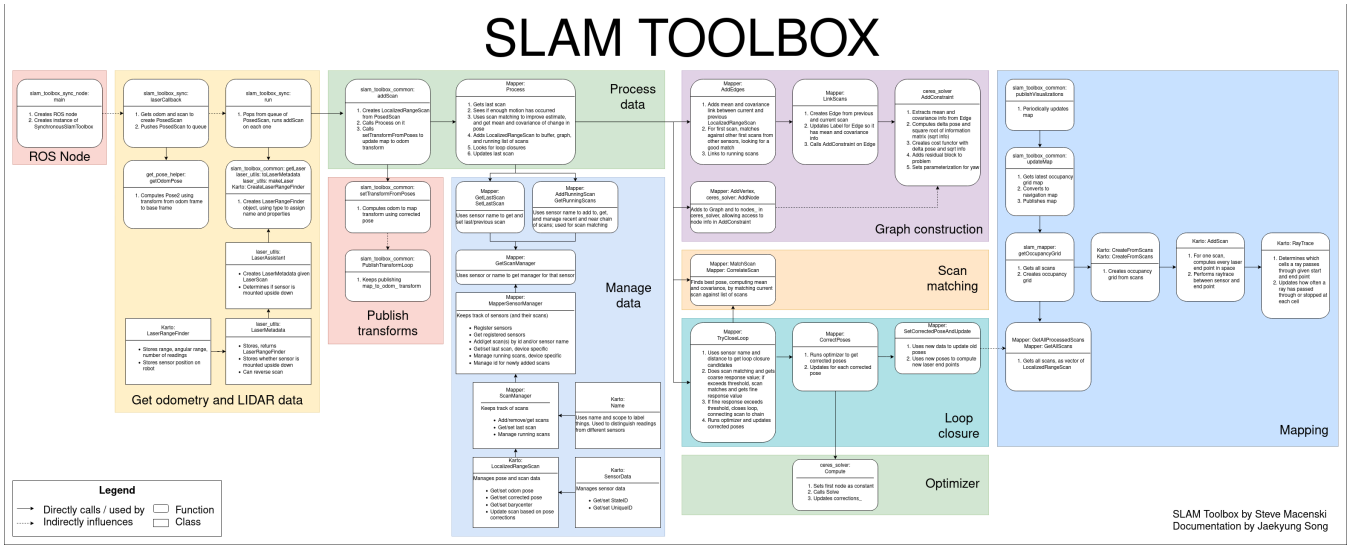
slam-toolbox is a ROS-based open-source 2D SLAM (simultaneous localization and mapping) algorithm package, mainly used for the mapping and localization of mobile robots in unknown environments. It is based on the **Graph Optimization** framework, combining sensor data such as LiDAR (e.g., 2D LiDAR) and an inertial measurement unit (IMU) to build a 2D grid map of the environment and update the robot's pose in real time.

2.1 Frontend Processing (Laser Scan Matching)

- **GICP algorithm**: estimates the robot's relative pose change by iteratively computing the optimal match between the current laser scan and the map. Compared with traditional ICP, GICP introduces a probabilistic model and is more robust to outliers.
- **Motion compensation**: uses IMU data to compensate for the laser scan distortion during the robot's motion, improving the matching accuracy in dynamic environments.

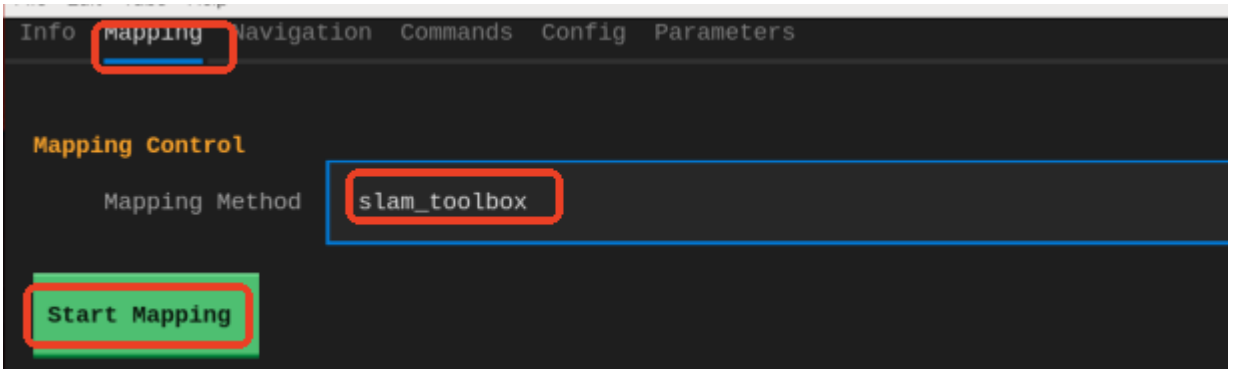
2.2 Backend Optimization (Graph Optimization)

- **Pose graph construction**: constructs a graph structure from the robot's pose nodes and the relative pose constraints obtained from scan matching.
- **Global optimization**: uses a nonlinear optimization algorithm (such as Ceres Solver) to optimize the pose graph, eliminating accumulated error and achieving global consistency of the map.



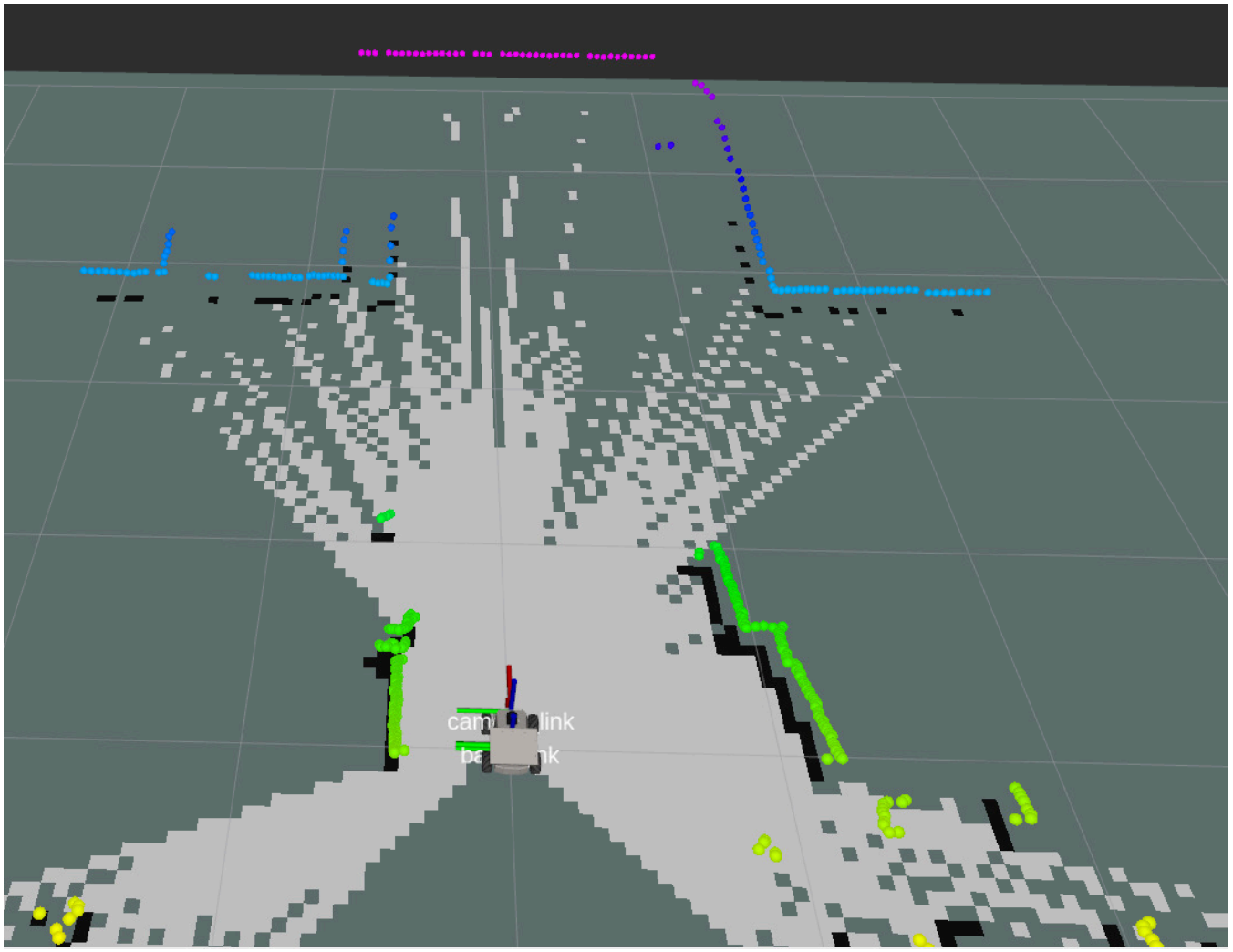
3. Running the Cases

- Launch method 1: GUI launch



- Launch method 2: CLI command launch in VNC

```
ros2 launch yahboomcar_bringup slam_toolbox.launch.py
```



- Control the car's movement. You can use keyboard control or handle control; here, keyboard control is used as an example

```
ros2 run common_demo keyboard_control
```

Click once on the terminal window with the mouse. Press z to appropriately reduce the speed

```
jetson@yahboom:~$ ros2 run common_demo keyboard_control

Reading from the keyboard  and Publishing to Twist!
---------------------------
Moving around:
   u    i    o
   j    k    l
   m    ,    .

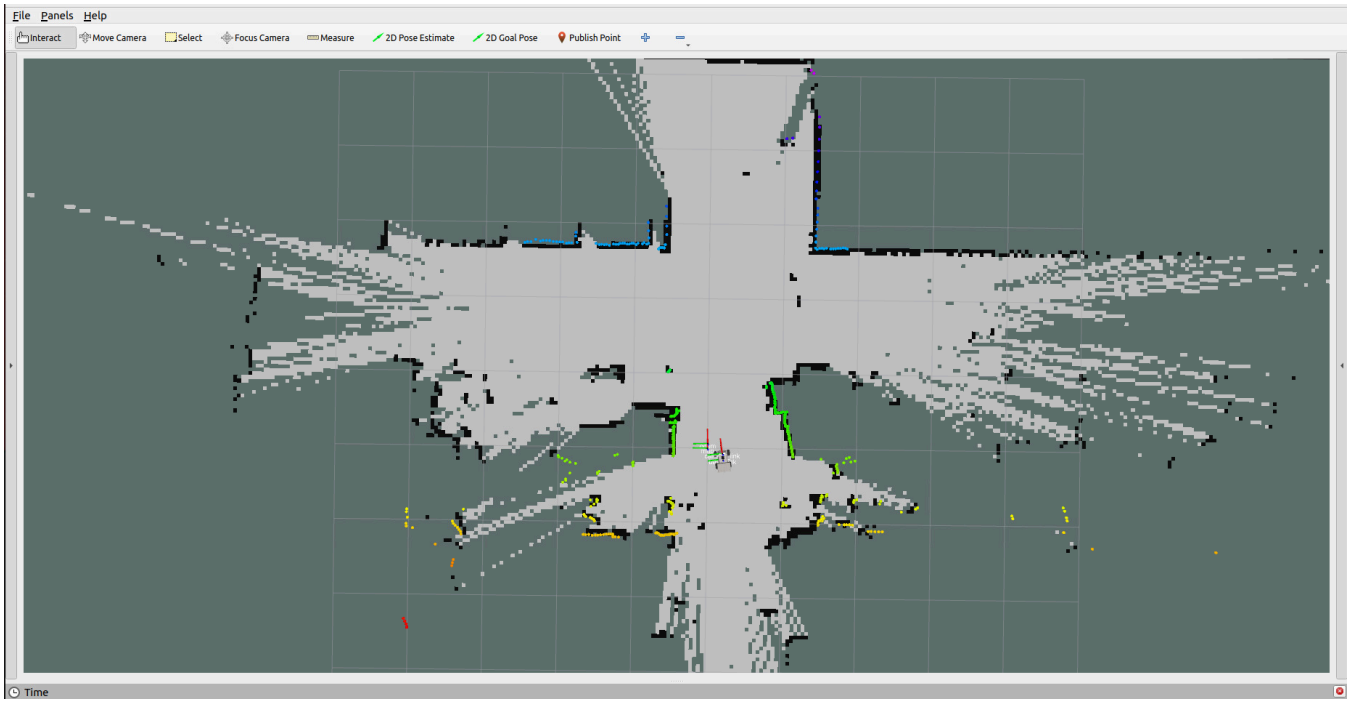
For Holonomic mode (strafing), hold down the shift key:
---------------------------
   U    I    O
   J    K    L
   M    <    >

t : up (+z)
b : down (-z)

anything else : stop

q/z : increase/decrease max speeds
w/x : increase/decrease only linear speed (step 0.02, range 0.01~2.0)
e/c : increase/decrease only angular speed (step 0.1, range 0.1~3.0)
```

- Press I, <, J, L to control the car's forward, backward, left turn, and right turn respectively, controlling the car to move slowly to complete mapping



3.1 Saving the Grid Map

- Start RobotManager

```
robotmanage gui
```

- Click Save Map. You can also customize the map name and map type. If the terminal indicates that saving failed, simply save again



- The map save path is as follows:

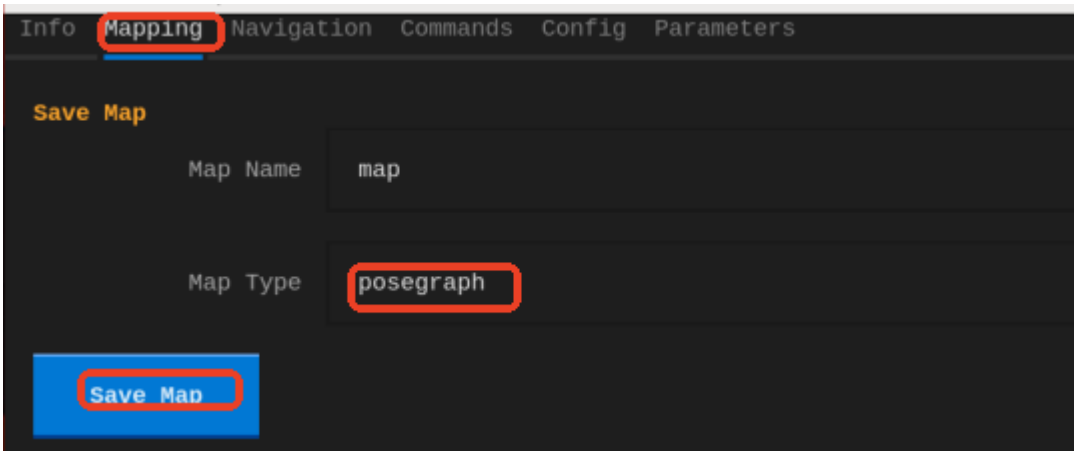
```
~/map/map.yaml
```

- A pgm image and a yaml file map.yaml

```
image: map.pgm
mode: trinary
resolution: 0.05
origin: [-10, -10, 0]
negate: 0
occupied_thresh: 0.65
free_thresh: 0.25
```

3.2 Saving the posegraph Pose Graph

- The pose graph can be used in localization mode during navigation



4. Configuration Parameters

- Parameter file path for synchronous mapping mode:

```
~/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_bringup/param/mapper_params_online_async.yaml
```

- See the comments for the role of each parameter

```
sync_slam_toolbox:
  ros__parameters:

    # Plugin parameters (optimization    solver_plugin: solver_plugins::CeresSolver  # Optimization solver plugin, using the Ceres solver
    ceres_linear_solver: SPARSE_NORMAL_CHOLESKY  # Ceres linear solver type, sparse normal Cholesky decomposition
    ceres_preconditioner: SCHUR_JACOBI  # Ceres preconditioner type, Schur-Jacobi preconditioner
    ceres_trust_strategy: LEVENBERG_MARQUARDT  # Ceres trust region strategy, Levenberg-Marquardt method
    ceres_dogleg_type: TRADITIONAL_DOGLEG  # Ceres dogleg type, traditional dogleg (takes effect only when the trust region strategy is DOGLEG)
    ceres_loss_function: None  # Ceres loss function, none (uses the default squared loss)

    # ROS parameters
    odom_frame: odom  # Name of the odometry coordinate frame
    map_frame: map  # Name of the map coordinate frame
    base_frame: base_footprint  # Name of the robot base coordinate frame
    scan_topic: /scan  # LiDAR scan data topic (absolute path)
    use_map_saver: true  # Whether to enable the map saving service and subscribe to the map topic
    mode: mapping  # Operating mode, mapping or localization

    # Map loading related parameters (if enabled, it will load the map from the specified file and continue mapping)
    map_file_name: ""  # The pose graph file name to load (if empty, nothing is loaded)
    # map_start_pose: [0.0, 0.0, 0.0]  # Starting pose [x, y, yaw] when loading the map (mutually exclusive with map_start_at_dock; this parameter takes priority)
```

```
    map_start_at_dock: true  # Whether to start loading from the first node of the map (the dock position)

    debug_logging: false  # Whether to enable debug log output
    throttle_scans: 1  # Scan data throttling count in synchronous mode (process 1 frame every n frames)
    transform_publish_period: 0.02  # Publishing period of the map-to-odometry coordinate transform (seconds); 0 means do not publish
    map_update_interval: 5.0  # Update interval of the 2D occupancy grid map (seconds, used for visualization or other applications)
    resolution: 0.05  # Resolution of the 2D occupancy grid map (meters/pixel)
    min_laser_range: 0.05  # Minimum valid laser distance used for grid map generation (meters)
    max_laser_range: 12.0  # Maximum valid laser distance used for grid map generation (meters)
    minimum_time_interval: 0.5  # Minimum time interval for processing two frames of scan data in synchronous mode (seconds)
    transform_timeout: 0.2  # Timeout for TF coordinate transform lookup (seconds)
    tf_buffer_duration: 30.  # Cache duration of TF messages (seconds)
    stack_size_to_use: 40000000  # Program stack size (bytes), used for serialization/deserialization of large maps
    enable_interactive_mode: true  # Whether to enable interactive mode (retains laser scan cache for RVIZ interaction, which increases memory usage)

    # General parameters
    use_scan_matching: true  # Whether to use scan matching
    use_scan_barycenter: true  # Whether to use the scan barycenter (improves matching stability)
    minimum_travel_distance: 0.5  # Minimum distance the robot must move before processing new scan data (meters)
    minimum_travel_heading: 0.5  # Minimum angle the robot must turn before processing new scan data (radians)
    scan_buffer_size: 10  # Scan data buffer size (keeps the most recent n scans)
    scan_buffer_maximum_scan_distance: 10.0  # Maximum distance of laser points in the scan buffer (meters; beyond this they are filtered out)
    link_match_minimum_response_fine: 0.1  # Minimum response value for link matching in fine matching (used for adjacent node constraints)
    link_scan_maximum_distance: 1.5  # Maximum distance between adjacent scans (meters, used to build links between nodes)
    loop_search_maximum_distance: 3.0  # Maximum search distance for loop closure detection (meters)
    do_loop_closing: true  # Whether to enable loop closure detection
    loop_match_minimum_chain_size: 10  # Minimum node chain length required for loop closure detection (to avoid mismatches caused by short chains)
    loop_match_maximum_variance_coarse: 3.0  # Maximum variance threshold during coarse matching (used to filter candidate loop closures)
    loop_match_minimum_response_coarse: 0.35  # Minimum response value for loop closures in coarse matching
    loop_match_minimum_response_fine: 0.45  # Minimum response value for loop closures in fine matching

    # Correlation parameters - scan matching parameters
```

```
    correlation_search_space_dimension: 0.5  # Search space size for scan matching (meters; the optimal match is searched within this range)
    correlation_search_space_resolution: 0.01  # Resolution of the scan matching search space (meters; affects matching accuracy and speed)
    correlation_search_space_smear_deviation: 0.1  # Smear deviation of laser points in scan matching (meters; increases robustness)

    # Correlation parameters - loop closure detection parameters
    loop_search_space_dimension: 8.0  # Search space size for loop closure detection (meters)
    loop_search_space_resolution: 0.05  # Resolution of the loop closure detection search space (meters)
    loop_search_space_smear_deviation: 0.03  # Smear deviation of laser points in loop closure detection (meters)

    # Scan matcher parameters
    distance_variance_penalty: 0.5  # Distance variance penalty coefficient (affects the weight of distance error in matching)
    angle_variance_penalty: 1.0  # Angle variance penalty coefficient (affects the weight of angle error in matching)

    fine_search_angle_offset: 0.00349  # Angular offset for fine search (radians, about 0.2 degrees)
    coarse_search_angle_offset: 0.349  # Angular offset for coarse search (radians, about 20 degrees)
    coarse_angle_resolution: 0.0349  # Angular resolution for coarse search (radians, about 2 degrees)
    minimum_angle_penalty: 0.9  # Minimum angle penalty coefficient (filters out matches with excessive angle deviation)
    minimum_distance_penalty: 0.5  # Minimum distance penalty coefficient (filters out matches with excessive distance deviation)
    use_response_expansion: true  # Whether to use response expansion (improves matching stability)
    min_pass_through: 2  # Minimum number of points passing through in scan matching (filters out invalid matches)
    occupancy_threshold: 0.1  # Threshold of the occupancy grid (used to determine whether a grid cell is occupied)
```